

Python FullStack Development

From Frontend to Backend

Web Development

What is internet ?

What is server ?

RBAC

1. Local Host
2. web Hosting

- Full stack developer
- Frontend Developer
- Backend Developer

Client - Server Architecture

<https://www.nslookup.io/>

DataBase

Python fullstack / Mern stack / Java full stack / Dot net full stack

Purpose of .html, .css and .js

1. How web applications works internally ?
2. What exactly happens when you search for the website inside a browser ?

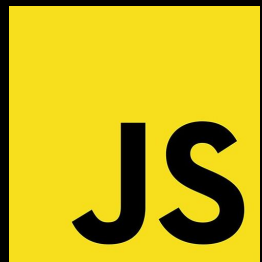
Frontend Technologies



HTML



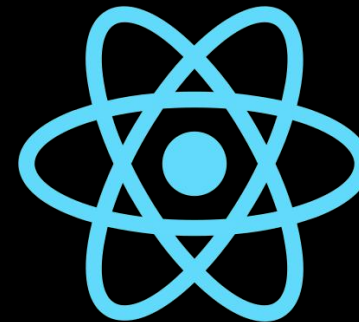
CSS



Js



Bootstrap



React

HTML (Hypertext Markup Language)

- What is HTML?
- Structure Tags
- Boilerplate code
- HTML Tags

`<h1> to <h6> , <p> , <a> , , <div> , ,
 , , <form> , <input> , <button> ,
 ,
 , <table> , <tr> , <th> , <td> , <blockquote> , <q>
<abbr> , <address> , <iframe>`

- HTML tables
- Types of HTML Tags
- Types of Attributes in HTML
- Tag / Element / Attribute
- HTML Comments
- HTML Favicon

CSS (Cascading Style Sheets)

What is CSS?

Why Use CSS?

Three Ways to Add CSS:

Benefits of CSS

CSS Syntax

Text properties : values

CSS Selectors

1. Universal Selector
2. Element Selector
3. ID Selector
4. Class Selector
5. Grouped Selector
6. Descendant & Child Selectors
7. Attribute Selectors
8. Pseudo-classes
9. Pseudo-elements

CSS Box Model

1. Margin
2. Padding
3. Border

display property

1. inline
2. block
3. inline-block

Image float

position property

1. static
2. relative
3. absolute
4. fixed
5. sticky

Pseudo-classes in CSS

1. :active
2. :hover
3. :focus

CSS Transforms

CSS Transitions

CSS Animations

Responsive Web design

1. Media Queries
2. Flexbox
3. Grid Layout
4. Bootstarp layout system

Bootstrap

What is Bootstrap?

How to Include Bootstrap ?

How Bootstrap working ?

Why Use Bootstrap?

Text Classes

Background Color Classes

Spacing Classes (Margin & Padding)

Flexbox Utility Classes

Width & Height Classes

Bootstrap Container

Bootstrap Grid System

JavaScript

What is JavaScript?

Js importance

History

How to add .Js to the HTML

JavaScript Engine and Execution Process

Variables

Datatypes (primitive, Non-Primitive)

operators

control flow statements

Functions

Hoisting & Temporal Dead Zone (TDZ)

- Types of functions
- Function Declaration
 - Function Expression
 - Arrow Functions (ES6)
 - Anonymous Functions
 - Callback Functions
 - Recursive Functions
 - Higher-Order Functions

Scope of a variable

setTimeout(), setInterval

clousers

Excecution Context and Call stack

Error Handling

BOM and DOM

Form data Handling

Installing **node**

Oops in Js

Events in js

APIs

js - synchronized - Asynchronized

- Callbacks
- Promises
- async and await
- Fetch Method
- Axios

React.js

- What is React
- The story
- Features of React.js
- Vite + React: Local Environment Setup
- Folder Structure (After Setup)
- JSX (JavaScript XML)
- React Components
- React Props
- import & export in React (ES6 Modules)
- Styling in React
- Rendering Arrays in React
- Conditional Rendering in React
- Events in React.js
- State, render and rerender
- state and Lifting State Up in React

Hooks in React

- useState
- useEffect
- useRef
- useContext
- useMemo
- useCallback

- Class-Based Components in React
- Forms in React
- Custom Hooks in React
- Higher-Order Component (HOC)
- Routing in Web Development
- Lazy Loading in React
- Performance Optimizing Techniques

Backend Technologies



Python

django



SQL

Python

- **What is python ?**
- **characteristics of python ?**
- **REPL in Python**
- **Variable in Python**
- **Data Types in Python**
 1. Primitive
 2. Non-Primitive
- **Operators in Python**
- **Control Flow Statements**
 1. Conditional Statements (Decision Making)
 2. Looping Statements (Repetition)
 3. Jumping Statements (Control Transfer)
- **Comprehensions in Python**
 - **1. List Comprehension**
 - **2. Dictionary Comprehension**
 - **3. Set Comprehension**
 - **4. Generator Comprehension**
- **Functions in Python**
 1. Parameters / Arguments
 2. Types of Function Arguments
 3. Scope in Python
 4. Nested Function
- **Types of Functions in Python**
 - 1. Built-in functions
 - 2. User-defined functions
 - 3. Anonymous (Lambda) Function
 - 4. Recursive Function
 - 5. Higher-Order Function (HOF)
 - 6. Generator functions
 - 7. Closures and nested (inner) functions

Python

- **Programming Paradigms**
- **OOP (Object-Oriented Programming)**
- Class, Object and constructor
- Types of variables in oops
- Types of methods in oops
- **magic methods**
- **Inheritance**
- Types of inheritance
- **MRO (Method Resolution Order)**
- **Polymorphism**
- **Types of polymorphism**
- **Access Modifiers in Python**
- **Encapsulation**
- **Abstraction**
- **Exception Handling**
- **Modular Programming**
- **File Handling in Python**
- **Python database connectivity(PDBC)**
- **Regular Expressions**

SQL

- Introduction to Databases & DBMS
- What is SQL & Types of SQL Commands
- Installing MySQL & Tools (Workbench / CLI)
- Creating Database & Tables (CREATE)
- Data Types in SQL
- CRUD Operations (INSERT, SELECT, UPDATE, DELETE)
- Filtering Data (WHERE, AND, OR, LIKE, IN, BETWEEN)
- Sorting & Limiting Data (ORDER BY, LIMIT)
- Aggregate Functions (COUNT, SUM, AVG, MIN, MAX)
- Grouping Data (GROUP BY, HAVING)
- Constraints (PRIMARY KEY, FOREIGN KEY, UNIQUE, NOT NULL)
- Joins (INNER, LEFT, RIGHT, FULL JOIN)
- Subqueries & Nested Queries
- Indexes & Performance Basics
- Views, Stored Procedures & Transactions (ACID Basics)

Django

1. Django Fundamentals (Basics)

- Introduction to Django & MVT Architecture
- Installing Django & Project Setup
- Project vs App Structure
- Running Development Server
- Django File Structure Explanation
- URLs & Routing (urls.py)
- Views (Function-Based Views - FBV)
- Class-Based Views (CBV)
- Templates (HTML rendering)
- Template Tags & Filters
- Static Files (CSS, JS, Images)

2. Models & Database

- Models & ORM (Object Relational Mapping)
- Field Types in Models
- Migrations (makemigrations, migrate)
- Django Admin Panel
- CRUD Operations using ORM
- QuerySets & Filtering Data

3. Forms & User Input

- HTML Forms in Django
- Django Forms (forms.py)
- ModelForms
- Form Validation
- Widgets & Styling Forms
- Handling POST Requests
- CSRF Protection

Django

4. Django REST Framework

- Introduction to APIs & REST
- Installing DRF
- Serializers
- ModelSerializers
- API Views (APIView)
- ViewSets & Routers
- CRUD APIs with DRF

5. Authentication in APIs

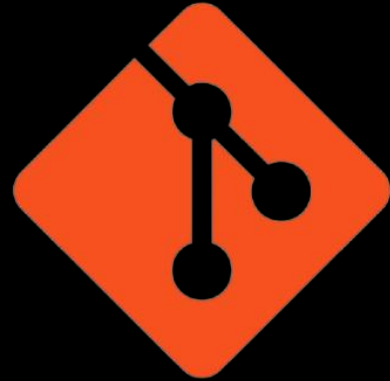
- Token Authentication
- JWT Authentication
- Login/Register APIs
- Permissions in DRF
- Role-Based APIs

Project : Job Portal

Build a Job Portal where recruiters post jobs and users apply (covers real-world use case).

- Use Django SSR for pages like job listings, job details, login/register using templates.
- Implement Models & ORM for Users, Jobs, Applications with relationships.
- Add Forms & Authentication (login, signup, role-based access: recruiter vs user).
- Create APIs using DRF for jobs & applications and connect with React (CSR).
- Integrate frontend using React + Axios + JWT Auth and deploy on VPS.

Version Control



Git



GitHub

Hosting Applications

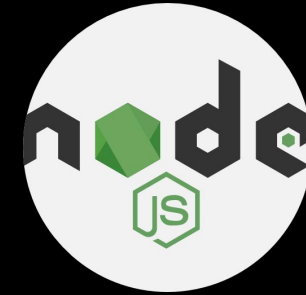
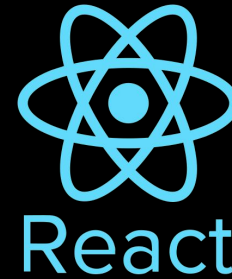
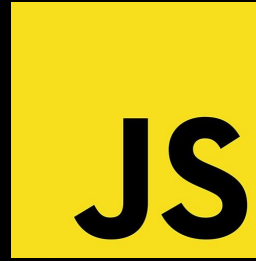


GitHub

Vercel



Tools & Technologies Covered



Vercel

DSA with Python

Basics of Programming & Complexity

- Python basics (lists, tuples, sets, dictionaries)
- Time & Space Complexity (Big-O, Big-Theta, Big-Omega)
- Best, Average, Worst case analysis
- Recursion basics
- Iterative vs Recursive approaches

Arrays & Strings

- Array operations (insert, delete, update, traverse)
- String manipulation
- Sliding Window technique
- Two Pointer approach
- Prefix Sum technique
- Kadane's Algorithm (Maximum Subarray)

Searching & Sorting

- Linear Search
- Binary Search (important for interviews)
- Bubble, Selection, Insertion Sort
- Merge Sort
- Quick Sort
- Counting Sort, Radix Sort (advanced basics)

Recursion & Backtracking

- Recursion tree understanding
- Base case & recursive case
- Backtracking concept
- Subsets, permutations
- N-Queens problem
- Sudoku Solver

DSA with Python

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Linked List

- Singly Linked List
- Doubly Linked List
- Circular Linked List
- Reverse a linked list
- Detect loop (Floyd Cycle Detection)
- Merge two linked lists

Stack & Queue

- Stack (LIFO)
- Queue (FIFO)
- Implementation using list & collections.deque
- Monotonic Stack
- Circular Queue
- Priority Queue (Heap basics)

Hashing

- HashMap (dictionary in Python)
- HashSet
- Collision handling
- Frequency counting
- Applications (two sum, duplicates)

DSA with Python

Trees

- Binary Tree basics
- Tree traversals (Inorder, Preorder, Postorder)
- Level Order Traversal (BFS)
- Height, Depth, Diameter
- Binary Search Tree (BST)
- Insert, delete, search in BST

Heaps / Priority Queue

- Min Heap & Max Heap
- Heap operations (insert, delete)
- Heapify
- Python heapq
- Top K elements problems

Graphs

- Graph representation (Adjacency List/Matrix)
- BFS (Breadth First Search)
- DFS (Depth First Search)
- Cycle detection
- Topological Sorting
- Shortest Path (Dijkstra)
- Minimum Spanning Tree (Prim's, Kruskal)

Greedy Algorithms

- Greedy approach concept
- Activity Selection
- Fractional Knapsack
- Huffman Coding
- Interval problems

DSA with Python

Dynamic Programming (DP)

- Memoization vs Tabulation
- Fibonacci DP
- 0/1 Knapsack
- Longest Common Subsequence (LCS)
- Longest Increasing Subsequence (LIS)
- DP on grids
- DP on subsequences

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